Reconstructing Constructivism: Causal Models and Bayesian Cognition (Gopnik & Wellman 2012)

Human children learn causal models of the world through hierarchical Bayesian inference and active counterfactual experimentation, contrasting sharply with pure correlational statistical text prediction.

GRADE: HIGH | STATUS: STABLE | DOMAIN: PHILOSOPHICAL, TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Children do not simply absorb passive statistical associations from environmental input; they construct abstract, coherent, causal representations ("theories") of physical, biological, and psychological domains. Gopnik & Wellman demonstrate that this cognitive development operates via hierarchical Bayesian model selection combined with active exploratory interventions (play and experimentation), generating genuine structural knowledge from sparse, noisy observations.

EVIDENCE AND CAUSAL ASSESSMENT

Laboratory experiments show that toddlers as young as two years old infer causal directionality, screen off spurious correlations, and design novel physical interventions after observing fewer than half a dozen statistical samples—a sample efficiency orders of magnitude beyond deep neural networks.

EFFECT SIZE AND PERSISTENCE

Where deep statistical models require billions of tokens to identify correlations that remain fragile to out-of-distribution shifts, human children acquire robust, transferable causal models that support counterfactual reasoning, explanation, and imaginative physical manipulation.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

Classified as robust (<20%). Causal inference mechanisms develop unevenly across domains (intuitive physics precedes complex theory of mind), but the Bayesian structural revision process is ubiquitous across cultures.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This does not imply that language models possess no utility for modeling human linguistic habits; it indicates that language models operate via passive correlational pattern matching rather than active, embodied causal world-modeling.

WHAT ACTION IS REASONABLY JUSTIFIED

Contrast LLM statistical sequence prediction with human causal reasoning when evaluating high-stakes decision systems; do not confuse fluent conversational mimicry with child-like causal understanding.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gopnik, A., & Wellman, H. M. (2012). Reconstructing constructivism: Causal models, Bayesian learning mechanisms, and the theory theory. Psychological Bulletin, 138(6), 1085–1108. https://doi.org/10.1037/a0028044